

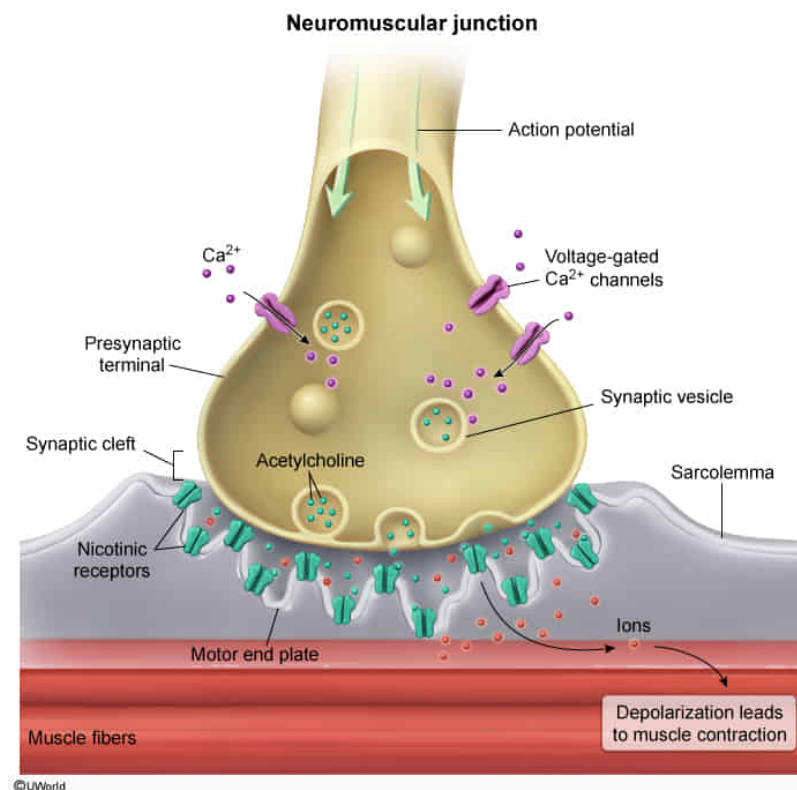
Impaired release of which of the following neurotransmitters would directly result in the paralysis of skeletal muscles?

- ☐ A. Norepinephrine [2%]
- ☐ B. Epinephrine [1%]
- ☐ C. Glutamate [6%]
- ✓ ☒ D. Acetylcholine [88%]

Correct

88% answered correctly

Explanation:



Acetylcholine (ACh), a neurotransmitter, plays an important role in the sympathetic and parasympathetic (autonomic) [nervous systems](#). In addition, ACh is the main neurotransmitter released by neurons of the efferent somatic nervous system, the system associated with skeletal muscle function.

Contraction of skeletal muscles is stimulated by transmission of a nerve impulse at the **neuromuscular junction**, the synapse between a skeletal muscle and a motor neuron. When an action potential reaches the terminal of a motor neuron, an influx of calcium through voltage-gated calcium channels triggers the release of ACh-containing vesicles (**exocytosis**) into the synaptic cleft. Released ACh binds to **receptors** of the motor end plates, depolarizing the innervated muscle fibers and causing **muscle contraction**. Consequently, disruption of the release of ACh inhibits muscle contractions and causes skeletal muscle paralysis.

(Choices A and B) Norepinephrine and epinephrine are amino acid–derived hormones released from the adrenal medulla as part of the fight-or-flight response. Norepinephrine is also the major postganglionic neurotransmitter of the sympathetic nervous system. These molecules are not directly involved in skeletal muscle contraction, so disruption of their release is unlikely to directly cause skeletal muscle paralysis.

(Choice C) Glutamate is the principal excitatory neurotransmitter of the central nervous system. It is not directly involved in neuromuscular transmission.

Educational objective:

At the neuromuscular junction, acetylcholine is released via exocytosis from presynaptic motor neurons. Acetylcholine binds to receptors on the motor end plate, triggering muscle contraction. Disruption of this process causes skeletal muscle paralysis.

Time spent:0 sec

Subject:

Biology

QID:401334

System:

Endocrine and Nervous Systems